UNIT 11

TRIGONOMETRY

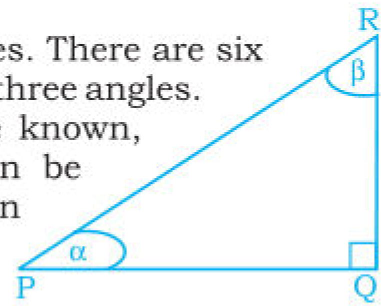
11.1 Trigonometry

11.1.1 Define Trigonometry

The word trigonometry is a Greek word. It is an important branch of Mathematics. Trigonometry deals with the relation of measures sides and angles of a triangle.

Many muslim mathematicians have made significant contribution. It plays an important role in the field of electronics, electrical engineering and many other branches of physical sciences.

Here we shall discuss right angled triangles. There are six elements of a triangle. i.e., Three sides and three angles. If the measures of any two elements are known, then measure of the other elements can be found. To find measures of the unknown elements of a right triangle, we use the knowledge of trigonometric ratios.

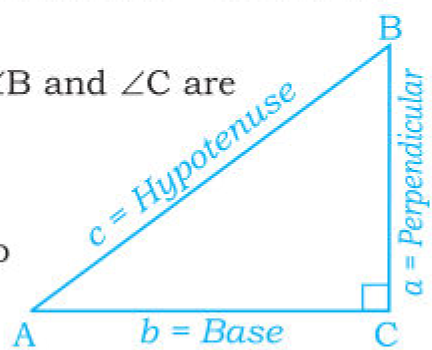


11.1.2 Trigonometric Ratios of Acute Angles

Consider a right angled triangle ACB in which $m\angle C = 90^\circ$, i.e., a right angle.

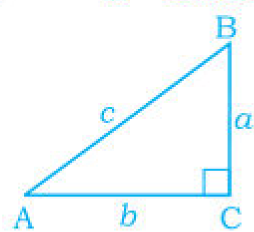
Measures of the sides opposite to $\angle A$, $\angle B$ and $\angle C$ are represented by a, b and c respectively,

i.e., $m\overline{BC} = a$, $m\overline{CA} = b$ and $m\overline{AB} = c$.



In right triangle, ratio of any of its two sides is called a trigonometric ratio.

Thus, we can form the following six possible trigonometric ratios. Consider an acute angle A of a right triangle ABC as under:



(i) Sine of $m\angle A$ = Sin $(m\angle A) = \dfrac{\text{Opp. Side of } m\angle A}{\text{Hypotenuse}} = \dfrac{m\overline{BC}}{m\overline{AB}} = \dfrac{a}{c}$

(ii) Cosine of $m\angle A$ = Cos $(m\angle A) = \dfrac{\text{Adj. Side of } m\angle A}{\text{Hypotenuse}} = \dfrac{m\overline{AC}}{m\overline{AB}} = \dfrac{b}{c}$

(iii) Tangent of $m\angle A$ = Tan $(m\angle A) = \dfrac{\text{Opp. Side of } m\angle A}{\text{Adjacent Side of } m\angle A} = \dfrac{m\overline{BC}}{m\overline{AC}} = \dfrac{a}{b}$

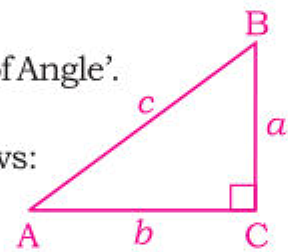
(iv) Cotangent of $m\angle A$ = Cot $(m\angle A) = \dfrac{\text{Adjacent Side of } m\angle A}{\text{Opposite Side of } m\angle A} = \dfrac{m\overline{AC}}{m\overline{BC}} = \dfrac{b}{a}$

(v) Secant of $m\angle A$ = Sec $(m\angle A) = \dfrac{\text{Hypotenuse}}{\text{Adjacent Side of } m\angle A} = \dfrac{m\overline{AB}}{m\overline{AC}} = \dfrac{c}{b}$

(vi) Cosecant of $m\angle A$ = Cosec $m\angle A = \dfrac{\text{Hypotenuse}}{\text{Opposite Side of } m\angle A} = \dfrac{m\overline{AB}}{m\overline{BC}} = \dfrac{c}{a}$

Note: For brevity in use;
(i) 'Side' instead of 'Measure of Side'
(ii) 'Trigonometric Ratio of Angle' instead of 'Measure of Angle'.

Activity: In the same way we can find the six Trigonometric ratios of another acute angle B as follows:



(i) Sine of $m\angle B$ = Sin $(m\angle B) = \dfrac{\text{Opp. Side of } \angle B}{\text{Hypotenuse}} = \dfrac{m\overline{AC}}{m\overline{AB}} = \underline{\quad\quad}$

(ii) Cosine of $m\angle B$ = Cos $(m\angle B) = \dfrac{\text{Adjacent Side of } m\angle B}{\text{Hypotenuse}} = \underline{\quad\quad} = \dfrac{a}{c}$

(iii) Tangent of $m\angle B$ = Tan $(m\angle B) = \dfrac{\text{Opp. Side of } m\angle B}{\underline{\quad\quad\quad\quad}} = \dfrac{m\overline{AC}}{m\overline{BC}} = \dfrac{b}{a}$

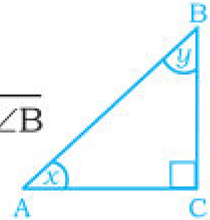
(iv) Cotangent of $m\angle B$ = Cot $(m\angle B) = \dfrac{\underline{\quad\quad\quad\quad}}{\text{Opposite Side of } m\angle B} = \dfrac{m\overline{BC}}{m\overline{AC}} = \underline{\quad\quad}$

(v) Secant of $m\angle B$ = Sec $(m\angle B) = \underline{\quad\quad\quad\quad} = \dfrac{m\overline{AB}}{m\overline{BC}} = \dfrac{c}{a}$

(vi) Cosecant of $m\angle B$ = Cosec $(m\angle B) = \dfrac{\text{Hypotenuse}}{\text{Opposite Side of } m\angle B} = \underline{\quad\quad} = \underline{\quad\quad}$

Note: Cosec θ is also denoted by Csc θ.

By considering the above ratios, it is observed that:



(i) $\text{Sin } m\angle A = \dfrac{1}{\text{Cosec } m\angle A}$ and $\text{Cosec } m\angle B = \dfrac{1}{\text{Sin } m\angle B}$

(ii) $\text{Cos } m\angle A = \dfrac{1}{\text{Sec } m\angle A}$ and $\text{Sec } m\angle B = \dfrac{1}{\text{Cos } m\angle B}$

(iii) $\text{Tan } m\angle A = \dfrac{1}{\text{Cot } m\angle A}$ and $\text{Cot } m\angle B = \dfrac{1}{\text{Tan } m\angle B}$

It is also observed that:

(a) $\dfrac{\text{Sin } m\angle A}{\text{Cos } m\angle A} = \text{Tan } m\angle A$ (b) $\dfrac{\text{Cos } m\angle B}{\text{Sin } m\angle B} = \text{Cot } m\angle B$

or $\dfrac{\text{Sin } x}{\text{Cos } x} = \text{Tan } x$ or $\dfrac{\text{Cos } y}{\text{Sin } y} = \text{Cot } y$

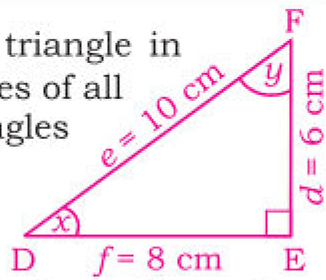
From the above results, we can get:

(iv) $\text{Sin } m\angle A \ \text{Cosec } m\angle A = 1$ or $\text{Sin } x \ \text{Cosec } x = 1$

(v) $\text{Cos } m\angle A \ \text{Sec } m\angle A = 1$ or $\text{Cos } x \ \text{Sec } x = 1$

(vi) $\text{Tan } m\angle A \ \text{Cot } m\angle A = 1$ or $\text{Tan } x \ \text{Cot } x = 1$

Example 1. Consider DEF, a right angled triangle in which $m\angle D = x$ and $m\angle F = y$. Find the values of all the trigonometric ratios, of both the acute angles of measure x and y.



Solution:

(i) $\text{Sin } x = \dfrac{\text{Opposite Side of } m\angle D}{\text{Hypotenuse}} = \dfrac{m\overline{EF}}{m\overline{DF}} = \dfrac{d}{e} = \dfrac{6}{10} = \dfrac{3}{5}$

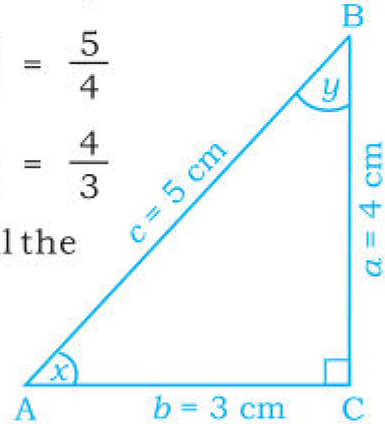
(ii) $\text{Cos } x = \dfrac{\text{Adjacent Side of } m\angle D}{\text{Hypotenuse}} = \dfrac{m\overline{DE}}{m\overline{DF}} = \dfrac{f}{e} = \dfrac{8}{10} = \dfrac{4}{5}$

(iii) $\text{Tan } x = \dfrac{\text{Opposite Side of } m\angle D}{\text{Adjacent Side}} = \dfrac{m\overline{EF}}{m\overline{DE}} = \dfrac{d}{f} = \dfrac{6}{8} = \dfrac{3}{4}$

(iv) $\text{Cosec } x = \dfrac{1}{\text{Sin } x} = \dfrac{e}{d} = \dfrac{10}{6} = \dfrac{5}{3}$

(v) $\text{Sec } x = \dfrac{1}{\text{Cos } x} = \dfrac{e}{f} = \dfrac{10}{8} = \dfrac{5}{4}$

(vi) $\text{Cot } x = \dfrac{1}{\text{Tan } x} = \dfrac{f}{d} = \dfrac{8}{6} = \dfrac{4}{3}$



Note: The students are advised to find all the trigonometric ratios of y themselves.

Example 2. In the given ΔABC, verify the following relations between trigonometric ratios.

(i) $\dfrac{\text{Cos } x}{\text{Sin } x} = \text{Cot } x$ **(ii)** $\dfrac{\text{Sin } y}{\text{Cos } y} = \text{Tan } y$ **(iii)** $\text{Tan } x \text{Cot } x = 1$

Solution: (i) $\dfrac{\text{Cos } x}{\text{Sin } x} = \text{Cot } x$

Here,

$$\text{L.H.S} = \frac{\text{Cos } x}{\text{Sin } x} = \frac{(m\overline{AC}/m\overline{AB})}{(m\overline{BC}/m\overline{AB})} = \frac{b/c}{a/c} = \frac{3/5}{4/5} = \frac{3}{\cancel{5}_1} \times \frac{\cancel{5}^1}{4} = \frac{3}{4} \quad \text{....I}$$

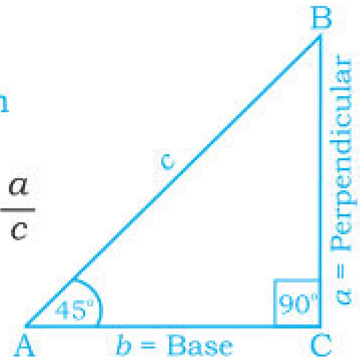
$$\text{R.H.S} = \text{Cot } x = \frac{\text{Adjacent Side of } m\angle A}{\text{Opposite Side of } m\angle A} = \frac{m\,\overline{AC}}{m\,\overline{BC}} = \frac{b}{a} = \frac{3}{4} \quad \text{....II}$$

Comparing I and II, we get L.H.S = R.H.S or $\dfrac{\text{Cos } x}{\text{Sin } x} = \text{Cot } x$

(ii) $\dfrac{\text{Sin } y}{\text{Cos } y} = \text{Tan } y$

$$\text{L.H.S} = \frac{\text{Sin } y}{\text{Cos } y} = \frac{(\text{Opposite Side of } m\angle B/\text{Hypotenuse})}{(\text{Adjacent Side of } m\angle B/\text{Hypotenuse})} = \frac{(m\overline{AC}/m\overline{AB})}{(m\overline{BC}/m\overline{AB})}$$

$$= \frac{b/c}{a/c} = \frac{3/5}{4/5} = \frac{3}{\cancel{5}_1} \times \frac{\cancel{5}^1}{4} = \frac{3}{4} \quad \text{...III}$$



Let us find the six trigonometric ratios of an acute angle of 45°.

(i) $\text{Sin } 45^\circ = \text{Sin } m\angle A = \dfrac{\text{Opp. Side of } m\angle A}{\text{Hyp}} = \dfrac{a}{c}$

$= \dfrac{b}{\sqrt{2}\,b} = \dfrac{1}{\sqrt{2}}$

Or $\text{Sin } 45^\circ = \dfrac{1}{\sqrt{2}}$

(ii) $\text{Cos } 45^\circ = \text{Cos } m\angle A = \dfrac{\text{Adj. Side of } m\angle A}{\text{Hypotenuse}} = \dfrac{b}{c} = \dfrac{b}{\sqrt{2}\,b} = \dfrac{1}{\sqrt{2}}$

or $\text{Cos } 45^\circ = \dfrac{1}{\sqrt{2}}$

(iii) $\text{Tan } 45^\circ = \text{Tan } m\angle A = \dfrac{\text{Opp. side of } m\angle A}{\text{Adj. side of } m\angle A} = \dfrac{a}{b} = 1$ or $\text{Tan } 45^\circ = 1$

(iv) $\text{Cot } 45^\circ = \text{Cot } m\angle A = \dfrac{\text{Adj. side of } m\angle A}{\text{Opp. side of } m\angle A} = \dfrac{b}{a} = 1$ or $\text{Cot } 45^\circ = 1$

(v) $\text{Sec } 45^\circ = \text{Sec } m\angle A = \dfrac{\text{Hyp}}{\text{Adjacent side of } m\angle A} = \dfrac{\sqrt{2}\,b}{b} = \sqrt{2}$

or $\text{Sec } 45^\circ = \sqrt{2}$

(vi) $\text{Cosec } m\angle A = \dfrac{\text{Hyp}}{\text{Opposite side of } m\angle A} = \dfrac{\sqrt{2}\,b}{b} = \sqrt{2}$

or $\text{Cosec } 45^\circ = \sqrt{2}$

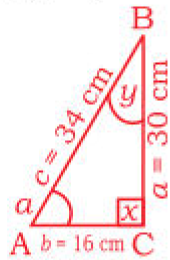
Thus

Sin 45°	Cos 45°	Tan 45°	Cot 45°	Sec 45°	Cosec 45°
$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$	1	1	$\sqrt{2}$	$\sqrt{2}$

(iii) $\text{Tan}\,\alpha.\,\text{Cot}\,\alpha = 1$ (iv) $\dfrac{\text{Sin}\,\beta}{\text{Cos}\,\beta} = \text{Tan}\,\beta$ (v) $\dfrac{\text{Cos}\,\alpha}{\text{Sin}\,\alpha} = \text{Cot}\,\alpha$

(vi) $\text{Cos}\,\alpha.\,\text{Cosec}\,\alpha = \text{Cot}\,\alpha$ (vii) $\dfrac{\text{Sec}\,\beta}{\text{Cosec}\,\beta} = \text{Tan}\,\beta$ (viii) $\dfrac{\text{Tan}\,\alpha}{\text{Cot}\,\alpha} = 1$

4. In the given right triangle ABC, Prove the following:



(i) $\text{Sin}\,x = \text{Cos}\,y$ (ii) $\text{Tan}\,y = \text{Cot}\,x$
(iii) $\text{Sec}\,x = \text{Cosec}\,y$ (iv) $\text{Cos}\,x = \text{Sin}\,y$
(v) $\text{Cot}\,y = \text{Tan}\,x$ (vi) $\text{Cosec}\,x = \text{Sec}\,y$

11.2 Trigonometric Ratios of Acute Angles

11.2.1 Find the values of all the trigonometric ratios of acute angles of 30°, 45°and 60° .

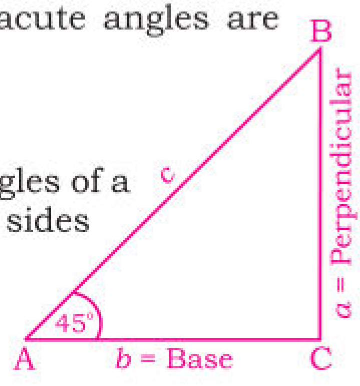
(a) Find the values of all the Trigonometric Ratios of an angle of measure of 45°.

Consider the right angled triangle ABC with right angle at point C and $m\angle A = 45^\circ$.

We know that in a right angled triangle, acute angles are complementary to each other.

So, $m\angle B = 90^\circ - m\angle A = 90^\circ - 45^\circ = 45^\circ$.

We know that if the measures of any two angles of a triangle are equal, then measures of the sides opposite to them are also equal.



i.e., $m\angle A = m\angle B = 45^\circ$, then $m\overline{BC} = m\overline{AC}$

We know that: $(\text{Hyp})^2 = (\text{Perp})^2 + (\text{Base})^2$

or $a = b$ (Pythagoras Theorem)

or $c^2 = a^2 + b^2$

or $c^2 = b^2 + b^2 = 2b^2$ $(\because\ a = b)$

$(c)^2 = (\sqrt{2b})^2$, i.e., $c = \sqrt{2}\ b$.

R.H.S= $\text{Tan } y = \dfrac{\text{Opposite Side of } m\angle B}{\text{Adjacent Side of } m\angle B} = \dfrac{m\overline{AC}}{m\overline{BC}} = \dfrac{b}{a} = \dfrac{3}{4}$ IV

Comparing results III and IV, we get L.H.S. = R.H.S

(iii) $\text{Tan } x \text{ Cot } x = 1$

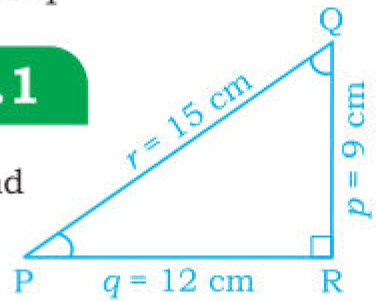
L.H.S = $\text{Tan } x \text{ Cot } x = \dfrac{\text{Opposite side of } m\angle A}{\text{Adjacent side of } m\angle A} \times \dfrac{\text{Adjacent side of } m\angle A}{\text{Opposite side of } m\angle A}$

$$\frac{a}{b} \times \frac{b}{a} = \frac{\cancel{4}^{1}}{\cancel{3}_{1}} \times \frac{\cancel{3}^{1}}{\cancel{4}_{1}} = \frac{1}{1} = 1$$

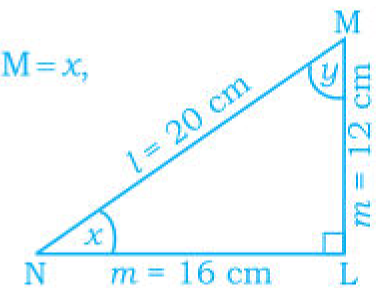
Thus L.H.S = R.H.S. Or $\text{Tan } x.\ \text{Cot } x = 1$

EXERCISE 11.1

1. In the given right angled triangle PQR find all the trigonometric ratios for both the acute angles



2. In the given right angled triangle LMN $m\angle M = x$, $m\angle N = y$. Find the values of the following trigonometric ratios.



(i) Sec x (ii) Cos y (iii) Tan x

(iv) Cosec y (v) Sec x (vi) Cot y

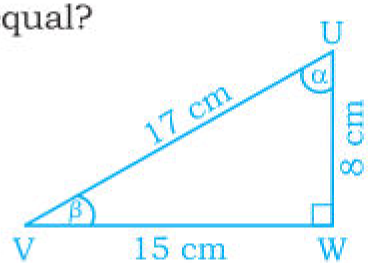
(vii) Cos x (viii) Sin y (ix) Cot x

(x) Sec y (xi) Cosec x (xii) Tan y

Which of the above trigonometric ratios are equal?

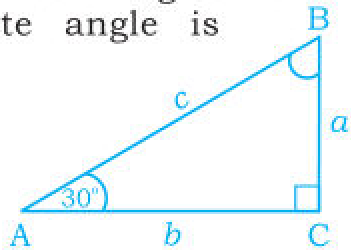
3. In the given right angled triangle UVW. Verify the following:

(i) $\text{Sin } \alpha . \text{Cosec } \alpha = 1$ (ii) $\text{Sec } \beta . \text{Cos } \beta = 1$



(B) To find the values of Trigonometric Ratios of an acute angle measuring 30°.

Consider the right angled triangle ABC, with right angle at point C and $m\angle A = 30^\circ$. In given right triangle, the acute angles are complementary angles, i.e., each acute angle is complement of the other.



So, $m\angle B = 90^\circ - m\angle A = 90^\circ - 30^\circ = 60^\circ$.

In the right triangle ABC, with $m\angle A = 30^\circ$, the measure of Hypotenuse is twice the measure of the side opposite to $\angle A$

i.e., $c = 2a$.

By Pythagoras Theorem, $c^2 = a^2 + b^2$

Putting $c = 2a$, we get: $(2a)^2 = a^2 + b^2$

or $4a^2 - a^2 = b^2$ or $3a^2 = b^2$ So, $b = \sqrt{3}\,a$

(i) $\text{Sin } 30^\circ = \dfrac{\text{Opposite Side of } m\angle A}{\text{Hypotenuse}} = \dfrac{m\overline{BC}}{m\overline{AB}} = \dfrac{a}{c} = \dfrac{a}{2a} = \dfrac{1}{2}$

(ii) $\text{Cos } 30^\circ = \dfrac{\text{Adjacent Side of } m\angle A}{\text{Hypotenuse}} = \dfrac{m\overline{AC}}{m\overline{AB}} = \dfrac{b}{c} = \dfrac{\sqrt{3}\,a}{2a} = \dfrac{\sqrt{3}}{2}$

(iii) $\text{Tan } 30^\circ = \dfrac{\text{Opposite Side of } m\angle A}{\text{Adjacent Side of } m\angle A} = \dfrac{m\overline{BC}}{m\overline{AC}} = \dfrac{a}{b} = \dfrac{a}{\sqrt{3}\,a} = \dfrac{1}{\sqrt{3}}$

(iv) $\text{Cot } 30^\circ = \dfrac{\text{Adjacent Side of } m\angle A}{\text{Hypotenuse}} = \dfrac{m\overline{AC}}{m\overline{BC}} = \dfrac{b}{a} = \dfrac{\sqrt{3}\,a}{a} = \sqrt{3}$

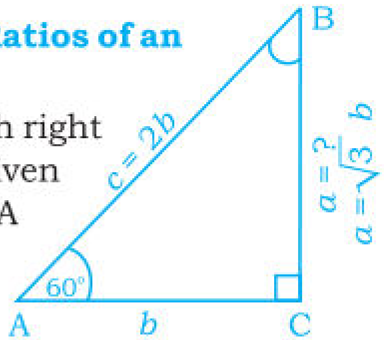
(v) $\text{Sec } 30^\circ = \dfrac{\text{Hypotenuse}}{\text{Adjacent Side of } m\angle A} = \dfrac{m\overline{AB}}{m\overline{AC}} = \dfrac{c}{b} = \dfrac{2\,a}{\sqrt{3}\,a} = \dfrac{2}{\sqrt{3}}$

(vi) $\text{Cosec } 30^\circ = \dfrac{\text{Hypotenuse}}{\text{Opposite Side of } m\angle A} = \dfrac{m\overline{AB}}{m\overline{BC}} = \dfrac{c}{a} = \dfrac{2\,a}{a} = 2$

Thus

Sin 30°	Cos 30°	Tan 30°	Cot 30°	Sec 30°	Cosec 30°
$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{3}}$	$\sqrt{3}$	$\frac{2}{\sqrt{3}}$	2

(c) To find the values of Trigonometric Ratios of an acute angle measuring 60°.



Consider the right angled triangle ABC, with right angle at point C and $m\angle A = 60^\circ$. In the given right angled triangle ABC, $m\angle B = 90^\circ - m\angle A = 90^\circ - 60^\circ = 30^\circ$. Thus, $m\angle A = 60^\circ$, $m\angle B = 30^\circ$ and $m\angle C = 90^\circ$,

Hence in the given right angled triangle ABC with $m\angle B = 30^\circ$, the measure of the Hypotenuse is twice the measure of the side opposite to it, i.e., $c = 2b$.

(By Pythagoras Theorem)

$(\text{Hypotenuse})^2 = (\text{Base})^2 + (\text{Perpendicular})^2$

$$c^2 = b^2 + a^2$$

or $\quad (2b)^2 = b^2 + a^2$ (putting $c = 2b$)

or $\quad 4b^2 - b^2 = a^2$

$\Rightarrow \quad a^2 = 3b^2 \quad$ or $\quad \sqrt{a^2} = \sqrt{3b^2}$

or $\quad a = \sqrt{3}\, b$

(i) $\text{Sin } m\angle A = \text{Sin } 60^\circ = \dfrac{\text{Opposite Side of } m\angle A}{\text{Hypotenuse}} = \dfrac{a}{c} = \dfrac{\sqrt{3}\, b}{2b} = \dfrac{\sqrt{3}}{2}$

(ii) $\text{Cos } m\angle A = \text{Cos } 60^\circ = \dfrac{\text{Adjacent Side of } m\angle A}{\text{Hypotenuse}} = \dfrac{b}{2b} = \dfrac{1}{2}$

(iii) $\text{Tan } m\angle A = \text{Tan } 60^\circ = \dfrac{\text{Opposite Side of } m\angle A}{\text{Adjacent Side of } m\angle A} = \dfrac{a}{b} = \dfrac{\sqrt{3}\, b}{b} = \sqrt{3}$

(iv) $\text{Cot } m\angle A = \text{Cot } 60^\circ = \dfrac{\text{Adjacent Side of } m\angle A}{\text{Opposite Side of } m\angle A} = \dfrac{b}{a} = \dfrac{b}{\sqrt{3}\, b} = \dfrac{1}{\sqrt{3}}$

(v) $\text{Sec } m\angle A = \text{Sec } 60^\circ = \dfrac{\text{Hypotenuse}}{\text{Adjacent Side of } m\angle A} = \dfrac{c}{b} = \dfrac{2\text{b}}{\text{b}} = 2$

(vi) $\text{Cosec } m\angle A = \text{Cosec } 60^\circ = \dfrac{\text{Hypotenuse}}{\text{Opp. Side of } m\angle A} = \dfrac{c}{a} = \dfrac{2\,b}{\sqrt{3}\, b} = \dfrac{2}{\sqrt{3}}$

Thus, the six trigonometric ratios for an angle measuring 60° are:

Sin 60°	Cos 60°	Tan 60°	Cot 60°	Sec 60°	Cosec 60°
$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$	$\frac{1}{\sqrt{3}}$	2	$\frac{2}{\sqrt{3}}$

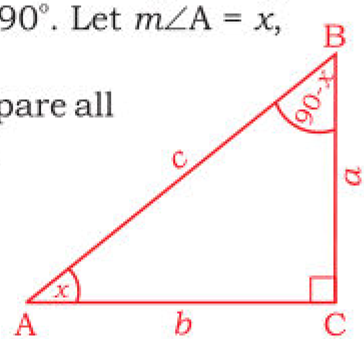
Note: Whatever be the size of a side in a right triangle, the trigonometric ratios of an acute angle is always the same.

The results derived above can be written in the form of a table:

Angle θ	Sin θ	Cos θ	Tan θ	Cot θ	Sec θ	Cosec θ
30°	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{3}}$	$\sqrt{3}$	$\frac{2}{\sqrt{3}}$	2
45°	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$	1	1	$\sqrt{2}$	$\sqrt{2}$
60°	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$	$\frac{1}{\sqrt{3}}$	2	$\frac{2}{\sqrt{3}}$

11.2.4 Define Trigonometric Ratios of Complementary Angles

We have already learnt that the acute angles of a right angled triangle are complementary angles. ABC is a right angled triangle, with $m\angle C = 90^\circ$, $m\angle A + m\angle B = 90^\circ$. Let $m\angle A = x$, then $m\angle B = 90^\circ - m\angle A$ or $m\angle B = 90^\circ - x$.



Consider right angled triangle ABC and compare all the six trigonometric ratios of $m\angle A = x$ and of $m\angle B = (90^\circ - x)$ which are complements of one another.

If $x = 30^\circ$, then angles of meaure 30° and $(90 - 30)^\circ$ or of 30° and 60° are complementary angles.

From the above figure, we see that:

$$\text{(i) Sin } x = \frac{a}{c} = \text{Cos }(90 - x) \Rightarrow \text{Sin } x = \text{Cos }(90 - x)$$

(ii) $\text{Cos } x = \frac{b}{c} = \text{Sin } (90 - x) \Rightarrow \text{Cos } x = \text{Sin } (90 - x)$

Also from the above table we see that:

$$\text{Sin } 30^\circ = \frac{1}{2} = \text{Cos } (90 - 30)^\circ \Rightarrow \text{Sin } 30^\circ = \text{Cos } 60^\circ$$

And $\text{Cos } 30^\circ = \frac{\sqrt{3}}{2} = \text{Sin } (90 - 30)^\circ \Rightarrow \text{Cos } 30^\circ = \text{Sin } 60^\circ$

From the above discussion, we conclude that:

(a) Sin and Cos of complementary angles are equal.

(b) Tan and Cot of complementary angles are equal.

Examples. $\text{Tan } 30^\circ = \frac{1}{\sqrt{3}} = \text{Cot } (90-30)^\circ \Rightarrow \text{Tan } 30^\circ = \text{Cot } 60^\circ$

$$\text{Cot } 30^\circ = \sqrt{3} = \text{Tan } (90-30)^\circ \Rightarrow \text{Cot } 30^\circ = \text{Tan } 60^\circ.$$

(c) Sec and Cosec of complementary angles are equal.

Examples. $\text{Sec } x = \text{Cosec } (90-x)$ and $\text{Cosec } x = \text{Sec } (90-x)$.

$$\text{Sec } 30^\circ = \frac{2}{\sqrt{3}} = \text{Cosec } (90-30)^\circ \Rightarrow \text{Sec } 30^\circ = \text{Cosec } 60^\circ.$$

$$\text{Sec } 60^\circ = 2 = \text{Cosec } (90-60)^\circ \Rightarrow \text{Sec } 60^\circ = \text{Cosec } 30^\circ.$$

Hence,

$$\text{Sin } 30^\circ = \frac{1}{2} = \text{Cos } 60^\circ$$

$$\text{Cos } 30^\circ = \frac{\sqrt{3}}{2} = \text{Sin } 60^\circ$$

$$\text{Tan } 30^\circ = \frac{1}{\sqrt{3}} = \text{Cot } 60^\circ$$

$$\text{Cot } 30^\circ = \sqrt{3} = \text{Tan } 60^\circ$$

$$\text{Sec } 30^\circ = \frac{2}{\sqrt{3}} = \text{Cosec } 60^\circ$$

$$\text{Cosec } 30^\circ = 2 = \text{Sec } 60^\circ$$

Similarly we can say:

$$\text{Sin } 40^\circ = \text{Cos } 50^\circ$$

$$\text{Cos } 70^\circ = \text{Sin } 20^\circ$$

$$\text{Tan } 10^\circ = \text{Cot } 80^\circ$$

$$\text{Cot } 20^\circ = \text{Tan } 70^\circ$$

$$\text{Sec } 50^\circ = \text{Cosec } 40^\circ$$

$$\text{Cosec } 20^\circ = \text{Sec } 70^\circ$$

Example. Find the value of the following:

(i) $\text{Sin } 30^\circ \times \text{Cos } 45^\circ + \text{Cos } 60^\circ \times \text{Sin } 45^\circ$

(ii) $\text{Tan } 30^\circ \times \text{Cot } 60^\circ + \text{Cot } 30^\circ \times \text{Tan } 60^\circ$

(iii) $\text{Sec } 30^\circ \times \text{Cosec } 45^\circ + \text{Sec } 45^\circ \times \text{Cosec } 60^\circ$

Solution:

(i) $\text{Sin } 30^\circ \times \text{Cos } 45^\circ + \text{Cos } 60^\circ \times \text{Sin } 45^\circ$

Putting the values from the table in the expression given above; we get

$\text{Sin } 30^\circ \times \text{Cos } 45^\circ + \text{Cos } 60^\circ \times \text{Sin } 45^\circ$

$$= \frac{1}{2} \times \frac{1}{\sqrt{2}} + \frac{1}{2} \times \frac{1}{\sqrt{2}}$$

$$= \frac{1}{2\sqrt{2}} + \frac{1}{2\sqrt{2}} = \frac{1+1}{2\sqrt{2}} = \frac{\cancel{2}}{\underset{1}{\cancel{2}}\sqrt{2}} = \frac{1}{\sqrt{2}}$$

(ii) $\text{Tan } 30^\circ \times \text{Cot } 60^\circ + \text{Cot } 30^\circ \times \text{Tan } 60^\circ$

$\text{Tan } 30^\circ = \frac{1}{\sqrt{3}}$, $\text{Tan } 60^\circ = \sqrt{3}$, $\text{Cot } 30^\circ = \sqrt{3}$

and $\text{Cot } 60^\circ = \frac{1}{\sqrt{3}}$ in the given expression,

$\text{Tan } 30^\circ \times \text{Cot } 60^\circ + \text{Cot } 30^\circ \times \text{Tan } 60^\circ$

$$= \frac{1}{\sqrt{3}} \times \frac{1}{\sqrt{3}} + \sqrt{3} \times \sqrt{3} = \frac{1}{\sqrt{3 \times 3}} + \sqrt{3 \times 3}$$

$$= \frac{1}{3} + \frac{3}{1} = \frac{1+9}{3} = \frac{10}{3} = 3\frac{1}{3} = 3.333$$

(iii) $\text{Sec } 30^\circ \times \text{Cosec } 45^\circ + \text{Sec } 45^\circ \times \text{Cosec } 60^\circ$

We know $\text{Sec } 30^\circ = \frac{2}{\sqrt{3}} = \text{Cosec } 60^\circ$ and $\text{Cosec } 45^\circ = \sqrt{2} = \text{Sec } 45^\circ$.

So, $\text{Sec } 30^\circ \times \text{Cosec } 45^\circ + \text{Sec } 45^\circ \times \text{Cosec } 60^\circ$

$$= \frac{2}{\sqrt{3}} \times \frac{2}{\sqrt{3}} + \frac{\sqrt{2} \times \sqrt{2}}{1} = \frac{2 \times 2}{\sqrt{3 \times 3}} + \frac{\sqrt{2 \times 2}}{1}$$

$$= \frac{4}{3} + \frac{2}{1} = \frac{4+6}{3} = \frac{10}{3} = 3\frac{1}{3} = 3.333$$

Example 2. Find: **(a)** Value of x when the value of $\text{Sin } x$ is $\frac{1}{2}$

(b) Value of y when the value of $\text{Tan } y = 1$

Solution: (a) We know that: $\text{Sin } 30^\circ = \frac{1}{2}$ (from the table)

But here $\text{Sin } x = \frac{1}{2}$

This implies that: $\text{Sin } x = \text{Sin } 30^\circ \Rightarrow x = 30^\circ$.

(b) We know that, Tan $45^\circ = 1$ (from the table)

But here Tan $y = 1 \Rightarrow \text{Tan } y = \text{Tan } 45^\circ \Rightarrow y = 45^\circ$.

EXERCISE 11.2

1. Find the value of x, when the value of Cos x is:

(i) $\frac{\sqrt{3}}{2}$ (ii) $\frac{1}{\sqrt{2}}$ (iii) $\frac{1}{2}$

2. Find the value of y, when the value of Tan y is:

(i) $\sqrt{3}$ (ii) 1 (iii) $\frac{1}{\sqrt{3}}$

3. Find the value of $m\angle A$, when the value of Cosec $m\angle A$ is:

(i) $\frac{2}{1}$ (ii) $\sqrt{2}$ (iii) $\frac{2}{\sqrt{3}}$

4. Find the value of the following:

(i) $\text{Sin } 30^\circ \times \text{Sin } 60^\circ + \text{Cos } 30^\circ \times \text{Cos } 60^\circ$

(ii) $\text{Sec } 30^\circ \times \text{Cosec } 60^\circ + \text{Tan } 60^\circ \times \text{Cot } 30^\circ$

(iii) $\text{Cos } 30^\circ \times \text{Sin } 30^\circ + \text{Cosec } 30^\circ \times \text{Sec } 30^\circ$

(iv) $\text{Sin } 60^\circ \times \text{Sin } 45^\circ - \text{Cos } 30^\circ \times \text{Cos } 45^\circ$

(v) $\text{Tan } 30^\circ \times \text{Cot } 60^\circ - \text{Sin } 45^\circ \times \text{Cos } 45^\circ$

(vi) $\text{Sec } 45^\circ \times \text{Cosec } 45^\circ + \text{Sec } 30^\circ \times \text{Cosec } 60^\circ$

(vii) $\text{Cot } 60^\circ \times \text{Sec } 30^\circ - \text{Cosec } 60^\circ \times \text{Tan } 30^\circ$

5. Find the value of x in each of the following cases:

(i) $\text{Sin } 30^\circ = \text{Cos } x$ (ii) $\text{Tan } 60^\circ = \text{Cot } x$ (iii) $\text{Sec } x = \text{Cosec } 45^\circ$

(iv) $\text{Cos } x = \text{Sin } 70^\circ$ (v) $\text{Sec } 40^\circ = \text{Cosec } x$ (vi) $\text{Cot } x = \text{Tan } 55^\circ$

(vii) $\text{Cosec } 10^\circ = \text{Sec } x$ (viii) $\text{Tan } x = \text{Cot } 65^\circ$ (ix) $\text{Sin } x = \text{Cos } 35^\circ$

6. Fill in the blanks:

(i) Sin 35° = Cos ____ (ii) Tan 65° = Cot ____ (iii) Sec 20° = Cosec ____

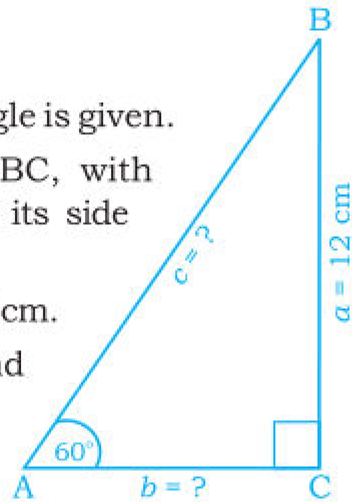
(iv) Cot 53° = Tan ____ (v) Cos 56° = Sin ____ (vi) Cosec 75° = Sec ____

11.2.5 Solve Right Angled Triangles using Trigonometric Ratios

We have already learnt that a triangle has six elements. If the measures of three of these elements, including at least one side are known, then measure of the other three elements of the triangle can be found. The process of finding the unknown elements of a triangle is known as **solving the triangle.**

Here we have to solve only right triangles.

Case I: Measure of one side and one acute angle is given.



Example 1. Solve right angled triangle ABC, with $m\angle C = 90°$, $m\angle A = 60°$ and the measure of its side opposite to $m\angle A = 12$ cm.

Solution: Here $m\angle C = 90°$, $m\angle A = 60°$ and $a = 12$cm.

Now we have to find $m\angle B$, $m\overline{AC} = b$ and $m\overline{AB} = c$

We also know that in a right angled triangle, acute angles are complementary.

Therefore, $m\angle A + m\angle B = 90°$

$$m\angle B = 90° - 60° = 30°$$

$$\text{Tan } m\angle A = \text{Tan } 60° = \frac{\text{Opposite Side of } m\angle A}{\text{Adjacent Side of } m\angle A} = \frac{a}{b} = \frac{12}{b}$$

or $\sqrt{3} = \frac{12}{b}$ or $b = \frac{12}{\sqrt{3}} = 4\sqrt{3} = 4 \times 1.732 = 6.928\text{cm}$

Again, $\text{Sin } m\angle A = \text{Sin } 60° = \frac{a}{c} = \frac{12}{c}$

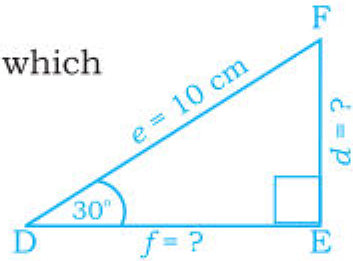
$\Rightarrow \frac{\sqrt{3}}{2} = \frac{12}{c}$ $\quad \Rightarrow c\sqrt{3} = 12 \times 2$

$$\Rightarrow c = \frac{24}{\sqrt{3}} = \frac{24\sqrt{3}}{\sqrt{3}\sqrt{3}} = \frac{\overset{1}{\cancel{24}}\sqrt{3}}{\underset{1}{\cancel{3}}} = 8\sqrt{3}$$

Thus, $m\angle B = 30°$, $m\overline{AC} = 4\sqrt{3} = 6.9$ cm, $m\overline{AB} = 13.8$ cm

Case II: Measure of Hypotenuse and one acute angle is given.

Example 2. Solve a right triangle DEF, when $m\angle E = 90^\circ$, $m\angle D = 30^\circ$ and $m\overline{DF}$ (Hypotenuse) = 10 cm.



Solution: Here DEF is a right triangle, in which $m\angle E = 90^\circ$, $m\angle D = 30^\circ$ and

Hypotenuse = mDF = 10 cm

We have to find $m\angle F$, $m\overline{DE} = f$, $m\overline{EF} = d$

We know that:

$$\text{Sin } m\angle D = \text{Sin } 30^\circ = \frac{\text{Opposite Side of } m\angle D}{\text{Hypotenuse}} = \frac{d}{e} = \frac{d}{10}$$

or $\quad \text{Sin } 30^\circ = \dfrac{d}{10}$

or $\quad \dfrac{1}{2} = \dfrac{d}{10}$, or $d = 5\text{cm}$

Also $\quad \dfrac{f}{e} = \text{Cos } m\angle D \Rightarrow \dfrac{f}{e} = \text{Cos } 30^\circ$

$$\Rightarrow \quad \frac{f}{10} = \frac{\sqrt{3}}{2} \quad \Rightarrow f = \frac{10\sqrt{3}}{2} = 5\sqrt{3} = 8.7 \text{ cm}$$

$$m\angle D + m\angle F = 90^\circ \text{ (Complementary angles)}$$

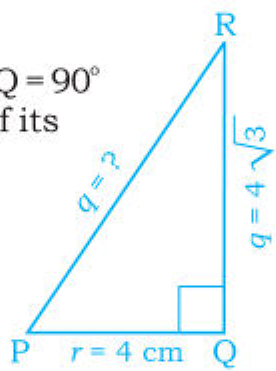
So, $\quad m\angle F = 90^\circ - m\angle D = 90^\circ - 30^\circ = 60^\circ$

Hence, the solution of the given ΔDEF is as under:

$$m\angle F = 60^\circ,\ m\overline{DE} = 5\sqrt{3} = 8.7 \text{ cm and },\ m\overline{EF} = 5 \text{ cm}$$

Case III: The measure of two sides is given.

Example 3. Solve a right angled triangle PQR with $m\angle Q = 90^\circ$ and measure of its perpendicular is $4\sqrt{3}$cm and of its base is 4cm.



Solution: Here ΔPQR is a right angled triangle, in which $m\angle Q = 90^\circ$, $m\overline{PQ} = 4$cm, $m\overline{QR} = 4\sqrt{3}$ cm.

Now we have to find $m\angle P$, $m\angle R$ and $m\overline{PR}$ = Hypothenuse.

According to Pythagoras Theorem:

$(\text{Hypotenuse})^2 = (\text{Base})^2 + (\text{Perpendicular})^2$

$$q^2 = r^2 + p^2$$

$$\Rightarrow (q)^2 = (4)^2 + (4\sqrt{3})^2 = 16 + 48 = 64$$

$$\therefore \quad \text{Hypotenuse} = q = \sqrt{64} = 8\text{cm}. \qquad \text{So,} \quad m\overline{PR} = 8 \text{ cm}.$$

We know that $\text{Tan } m\angle P = \dfrac{\text{Opposite Side of } m\angle P}{\text{Adjacent Side}} = \dfrac{4\sqrt{3}}{4} = \sqrt{3}$.

$$\therefore \quad m\angle P = 60^\circ$$

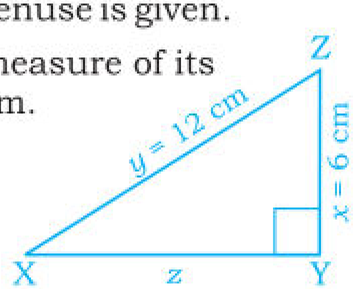
Then $m\angle R = 90^\circ - 60^\circ = 30^\circ$.

Hence solution of the given triangle is as under:

$$m\angle P = 60^\circ, \ m\angle R = 30^\circ \text{ and Hyp} = 8 \text{ cm}.$$

Case IV: The measures of one side and Hypotenuse is given.

Example 4. Solve ΔXYZ when $m\angle Y = 90^\circ$, measure of its Perpendicular is 6cm and Hypotenuse is 12 cm.



Solution:

Here we have to find the unknown elements.

$$m\angle X = ?, \ m\angle Z = ? \text{ and } m\overline{XY} = z = ?$$

We know that $\text{Sin } m\angle X = \dfrac{\text{Opposite Side of } m\angle X}{\text{Hypotenuse}} = \dfrac{m\overline{YZ}}{m\overline{XZ}} = \dfrac{6}{12} = \dfrac{1}{2}$.

Thus, $\text{Sin } m\angle X = \dfrac{1}{2} \Rightarrow m\angle X = 30^\circ$.

$$\text{Now } m\angle Z = 90^\circ - m\angle X = 90^\circ - 30^\circ = 60^\circ.$$

Now: $\dfrac{z}{y} = \text{Cos } m\angle X \quad \Rightarrow \dfrac{z}{12} = \text{Cos } 30^\circ = \dfrac{\sqrt{3}}{2}$

$$\Rightarrow z = \frac{\sqrt{3}}{2} \times 12 = 6\sqrt{3}$$

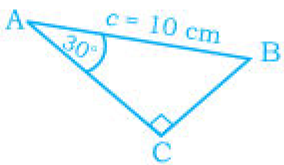
Hence, $m\angle X = 30^\circ$, $m\angle Z = 60^\circ$ and $m\overline{XY} = 6\sqrt{3}$ cm.

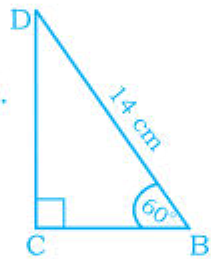
EXERCISE 11.3

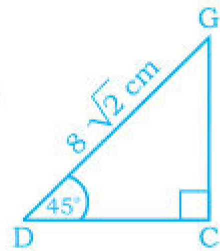
A. Solve the following right angled triangles:

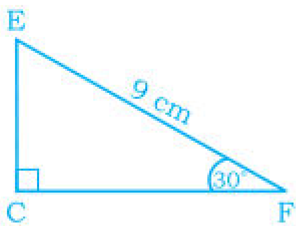
1. ΔABC, $m\angle C = 90^\circ$, $m\angle A = 30^\circ$ and Perpendicular $m\overline{BC} = 3cm$
2. ΔDEF, $m\angle E = 90^\circ$, $m\angle D = 60^\circ$ and Base $m\overline{DE} = 4cm$
3. ΔLMN, $m\angle M = 90^\circ$, $m\angle N = 45^\circ$ and Perpendicular $m\overline{ML} = 8cm$

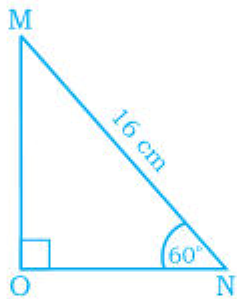
B. Find unknown elements of the following right angled triangles.

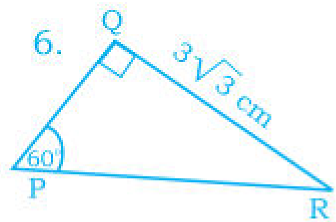
1. 

2. 

3. 

4. 

5. 

6. 

C. Solve the following right angled triangles.

1. ΔLMN, $m\angle M=90^\circ$, Perpendicular side $l = 4cm$ and Base side $n = 4\sqrt{3}$
2. ΔPQR, $m\angle Q=90^\circ$, Perpendicular $p = 6\sqrt{3}$ cm, and base side $r = 6cm$
3. ΔXYZ, $m\angle Y=90^\circ$, Perpendicular side $x = 4$ cm and Base side $z = 7$ cm.

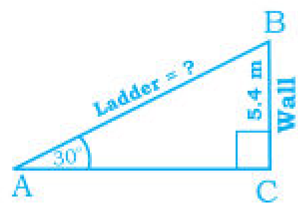
D. Solve the following right angled triangles.

1. ΔACB, $m\angle C = 90^\circ$, $a = 5cm$ and Hypotenuse $c = 10cm$
2. ΔDEF, $m\angle C = 90^\circ$, $d = 7\sqrt{3}$ cm and Hypotenuse $f = 14cm$
3. ΔLMN, $m\angle L = 90^\circ$, $m = 8cm$ and Hypotenuse $l = 8\sqrt{2}$ cm

11.2.6 To Solve Real Life Problems of Heights and Distances

The problems of finding heights and distances of objects in fact, is the same as that of solving right angled triangles. In this case we must draw the triangle. Sometimes, the triangle is a section of the diagram that accompanies the problem. However, we must redraw the triangle. This will avoid any confusion and help us not only to identify correctly the sides but also to choose the correct ratios.

Example 1. A window cleaner has a ladder. He leans it against a wall which makes an angle of 30° with the floor. It reaches the height of 5.4m on the wall. What is the length of the ladder?



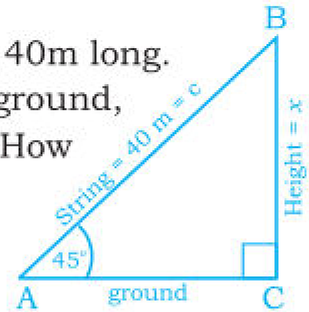
Solution: First we draw the situation as a right triangle.

Now $\frac{5.4}{c} = \text{Sin } m\angle A \Rightarrow \frac{5.4}{c} = \text{Sin } 30^\circ \Rightarrow \frac{5.4}{c} = \frac{1}{2}$

$\Rightarrow c \times 1 = 2 \times 5.4 \Rightarrow c = 10.8\text{m}$

Hence, length of the ladder is 10.8m.

Example 2. A boy is flying a kite with a string 40m long. The string is being held at 1m above the ground, making an angle of 45° with the horizontal. How high is the kite?



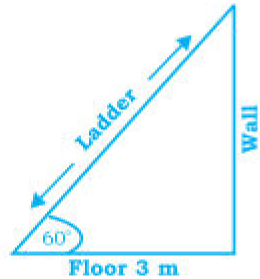
Solution: First we draw the figure which is a right triangle. Let x be the height of the kite.

Then $\frac{x}{c} = \frac{\text{Opposite Side of } m\angle A}{\text{Hypotenuse}} \Rightarrow \frac{x}{c} = \text{Sin } 45^\circ \Rightarrow \frac{x}{40} = \frac{1}{\sqrt{2}}$

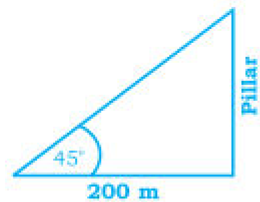
$\Rightarrow x = \frac{1}{\sqrt{2}} \times 40 = 0.707 \times 40 \Rightarrow x = 28.28 = 28.3 \text{ m (approx)}.$

Thus, kite is at a height of 28.3 m + 1 m = 29.3 m (approx) from the ground.

EXERCISE 11.3

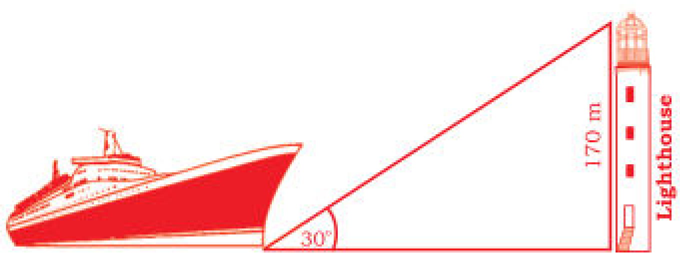


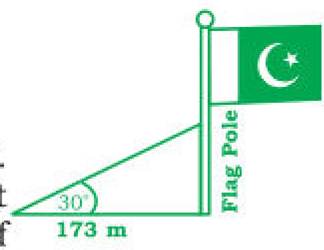
1. A ladder rests against a wall as shown in the figure. It makes an angle of 60° with the floor. If foot of the ladder is 3m away from the wall. What is the length of ladder?



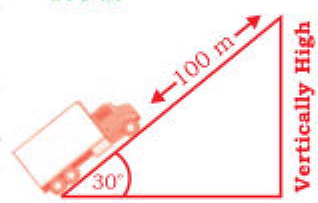
2. The foot of a pillar is at a distance of 200m from a point on the ground as shown in the figure. The angle which the pillar makes from this point is of 45°. Find the height of the pillar.

3. On the bank of a sea, there is a light house 170m high. A ship standing in the sea, makes an angle of 30° with the top of the light house, as shown in the figure. Find the distance between foot of the light house and the ship.

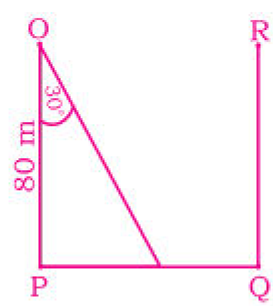




4. There is a flag-pole as shown in the figure. The angle of the top of the flag-pole from a point on the ground 20 metres away from the foot of the pole is 30°. Find the height of the pole.



5. A truck drives up a 100m incline. The angle that the truck drives up is of 30° as shown in the figure. How high did the truck rise vertically?



6. Two trees at points P and Q are on a side of a road. Tree $\overline{PO}$ is perpendicular on the road $\overline{PQ}$ as shown in the figure. Height $\overline{PO}$ is measured to be 80m. The angle POQ is measured as 30°. What is the horizontal distance between the trees?

REVIEW EXERCISE 11

1. Show that:

(i) $2\text{ Sin }45^\circ + \frac{1}{2}\text{ Cosec }45^\circ = \frac{3}{\sqrt{2}}$

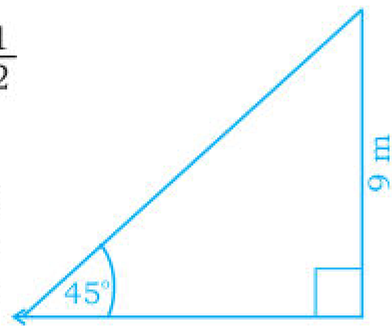
(ii) $\text{Cot }60^\circ . \text{ Cos }30^\circ - \text{Sin }60^\circ \text{ Sin }30^\circ = 0$

(iii) $\text{Sin }60^\circ \text{ Cos }30^\circ - \text{Cos }60^\circ \text{ Sin }30^\circ = \frac{1}{2}$

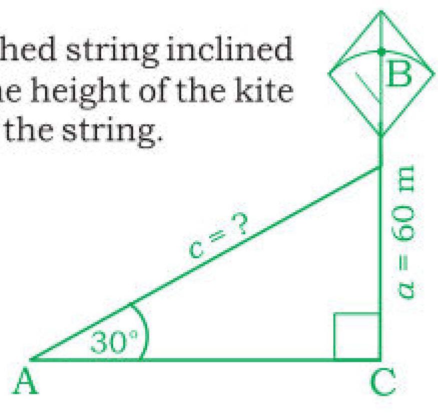
(iv) $(\text{Cos }30^\circ - \text{Sin }30^\circ) \times (\text{Cos }30^\circ + \text{Sin }30^\circ) = \frac{1}{2}$

(v) $\text{Sin }45^\circ \times \text{Cos }45^\circ + \text{Sin }30^\circ \times \text{Cos }60^\circ = \frac{3}{4}$

2. A tree having 9m height is on one bank of a river. It makes an angle of 45° on the opposite side of the river; as shown in the figure. Find the width of the river.



3. A girl flying a kite attached it to a stretched string inclined at an angle of 30° with the horizontal. If the height of the kite from the ground is 60m, find the length of the string.



4. Fill in the blanks:

(i) $\text{Cosec }30^\circ =$ ____________

(ii) $\text{Cot }60^\circ =$ ____________

(iii) $\text{Cos }45^\circ =$ ____________

(iv) $\text{Sec }30^\circ =$ ____________

(v) Reciprocal of $\text{Tan } m\angle C =$ ____________

(vi) $\text{Sin } m\angle A = \text{Cos}$ ____________

(vii) $\text{Cos }(90 - m\angle A) = \text{Sin}$ ____________

(viii) $\text{Tan }30^\circ = \text{Cot}$ ____________

(ix) $\text{Cosec}10^\circ = \text{Sin}$ ____________

(x) $\text{Sec }(90 - \theta) = \text{Cosec}$ ____________

5. Encircle the correct answer.

(i) The value of Cos 60° is ____________

(a) $\frac{\sqrt{3}}{2}$ (b) $\frac{1}{2}$ (c) $\frac{1}{\sqrt{2}}$

(ii) The value of Cot 30° = __________

(a) $\frac{1}{\sqrt{3}}$ (b) 1 (c) $\sqrt{3}$

(iii) The value of Tan 45° = __________

(a) 2 (b) $\sqrt{2}$ (c) 1

(iv) The value of $\text{Cos}^2\ 30°$ = __________

(a) $\frac{3}{4}$ (b) $\frac{\sqrt{3}}{2}$ (c) $\frac{1}{\sqrt{2}}$

(v) The value of Cosec 60° = __________

(a) $\frac{\sqrt{3}}{2}$ (b) $\frac{2}{\sqrt{3}}$ (c) $\frac{1}{2}$

(vi) The value of $\text{Sin}^2\ 45°$ = __________

(a) $\frac{1}{\sqrt{2}}$ (b) 2 (c) $\frac{1}{2}$

(vii) The value of $\text{Cosec}^2\ 60°$ = __________

(a) 2 (b) $\frac{4}{3}$ (c) $\frac{2}{\sqrt{3}}$

6. Write T for true and F for false.

(i) $\text{Cos}\,\theta = \frac{1}{\text{Sin}\,\theta}$ (ii) $\text{Cosec}\,\theta = \frac{1}{\text{Sec}\,\theta}$ (iii) $\sec\beta = \frac{1}{\text{Cosec}\,\beta}$

(iv) $\text{Tan}\,\theta.\ \text{Cot}\,\theta = 1$ (v) $\text{Tan}\,x.\ \text{Cot}\,x = 1$ (vi) $\frac{\text{Adj. Side of } m\angle A}{\text{Hypotenuse}} = \text{Sin}\ m\angle A$

(vii) $\text{Cot}\,x = \frac{\text{Cos}\,x}{\text{Sin}\,x}$ (viii) Sin 20° = Cosec 70°

(ix) The inverse of Sin x = Cosec x (x) Cos (90° – 30°) = Sec 60°

SUMMARY

- Trigonometry is the relation of measurement of elements of a triangle.
- There are six trigonometric ratios; Viz, Sine, Cosine, Tangent, Cotangent, Secant and Cosecant.
- Trigonometric ratios are used to relate the measures of angles to the lengths of the sides of a right triangle.
- Three most common ratios in trigonometry are:

(i) $\text{Sin}\,\theta = \dfrac{\text{Opposite Side of given angle}}{\text{Hypotenuse}}$

(ii) $\text{Cos}\,\theta = \dfrac{\text{Adjacent Side of given angle}}{\text{Hypotenuse}}$

(iii) $\text{Tan}\,\theta = \dfrac{\text{Opposite Side of given angle}}{\text{Adjacent Side of given angle}}$

- $\text{Sin}\,\theta = \dfrac{1}{\text{Cosec}\,\theta}$, $\text{Cos}\,\theta = \dfrac{1}{\text{Sec}\,\theta}$, $\text{Tan}\,\theta = \dfrac{1}{\text{Cot}\,\theta}$
- $\text{Tan}\,(90^\circ - \theta) = \text{Cot}\,\theta$, $\text{Cot}\,(90^\circ - \theta) = \text{Tan}\,\theta$
- $\text{Sec}\,(90^\circ - \theta) = \text{Cosec}\,\theta$, $\text{Cosec}\,(90^\circ - \theta) = \text{Sec}\,\theta$

Table for values of Trigonometric ratios of angles of different measures.

θ	Sin	Cos	Tan	Cot	Sec	Cosec
30°	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{3}}$	$\sqrt{3}$	$\frac{2}{\sqrt{3}}$	2
45°	$\frac{1}{\sqrt{2}}$	$\frac{1}{\sqrt{2}}$	1	1	$\sqrt{2}$	$\sqrt{2}$
60°	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$	$\sqrt{3}$	$\frac{1}{\sqrt{3}}$	2	$\frac{2}{\sqrt{3}}$